

Foundation (Jalapeno)

- Q1.** The quadratic equation $x^2 + 4x + 4 = 0$ has how many solutions?
(A) 0
(B) 1
(C) 2
(D) 4
(E) 6
- Q2.** What is the discriminant of the quadratic equation $x^2 - 7x + 12 = 0$?
(A) -5
(B) 1
(C) 13
(D) 25
(E) 37
- Q3.** The quadratic equation $2x^2 - 5x - 3 = 0$ represents the trajectory of a spaceship. How many times will it cross the x-axis?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q4.** To find the time it takes for a planet to orbit the Sun, scientists use the equation $t^2 - 10t + 16 = 0$. What is the discriminant?
(A) 36
(B) 48
(C) 64
(D) 100
(E) 144
- Q5.** The quadratic equation $x^2 - 9x + 20 = 0$ represents the path of a comet. What are the solutions to this equation?
(A) $x = 4, x = 5$
(B) $x = 2, x = 10$
(C) $x = 1, x = 20$
(D) $x = 5, x = 4$
(E) $x = 0, x = 20$
- Q6.** A lab experiment involves a quadratic equation $d^2 + 2d - 6 = 0$. What can be said about the solutions of this equation?
(A) Two distinct real solutions
(B) One real solution
(C) No real solutions
(D) Infinitely many solutions
(E) Two complex solutions
- Q7.** To calculate the speed of a spaceship, the equation $v^2 - 4v - 21 = 0$ is used. What is the nature of the solutions?
(A) One real and one complex solution
(B) Two distinct real solutions
(C) Two equal real solutions
(D) No real solutions
(E) Infinitely many solutions
- Q8.** The equation $2x^2 + 5x + 3 = 0$ represents the gravitational force between two planets. What is the discriminant of this equation?
(A) 13
(B) 17
(C) 25
(D) 37
(E) 41

Open Ended Questions (FRQ)

- Q9.** A spacecraft is traveling through space with a velocity given by $v^2 - 4v - 5 = 0$. Solve for v .
- Q10.** The quadratic equation $x^2 - 3x - 18 = 0$ represents the distance traveled by a rocket. Find the solutions and state the nature of the solutions.

Chef's Tips

- Emphasize understanding of the quadratic formula and its application.
- Use real-world examples from space and science to demonstrate the relevance of the concept.
- Encourage students to check their solutions by plugging them back into the original equation.